

Retail Customer Behavior Analytics

Project Report — Data Preparation, SQL Analysis & Power BI Dashboard

Python

PostgreSQL

Power BI

1. Project Overview

This project examines shopping behavior across 3,900 retail transactions to understand what drives customer spending, product preference, and subscription loyalty. The goal is to turn raw transactional data into a set of actionable insights that a retail management team could use to shape marketing, discounting, and product strategy.

2. Dataset Summary

3,900

Transactions

18

Columns

37

Missing review ratings

4

Product categories

Key fields fall into three groups:

- **Customer demographics** — age, gender, location, subscription status
- **Purchase details** — item purchased, category, purchase amount, season, size, color
- **Shopping behavior** — discount applied, promo code used, previous purchases, purchase frequency, review rating, shipping type

3. Data Preparation (Python)

The raw dataset was cleaned and transformed in `pandas` before being loaded into PostgreSQL:

- **Missing data handling** — the 37 missing values in `review_rating` were imputed with each product category's median rating, rather than a single global median, to keep the imputation category-aware.
- **Column standardization** — all column names were converted to `snake_case` for consistency between Python, SQL, and Power BI.
- **Feature engineering** — an `age_group` column (Young Adult / Adult / Middle-aged / Senior) was created via quartile binning, and a `purchase_frequency_days` column translated categorical purchase frequency into a numeric day count.
- **Redundancy check** — `discount_applied` and `promo_code_used` were found to be identical across every row, so `promo_code_used` was dropped.
- **Database load** — the cleaned DataFrame was loaded into PostgreSQL via SQLAlchemy for structured

4. SQL Analysis — Key Findings

Ten business questions were answered directly in PostgreSQL.

1. Revenue by Gender

Gender	Revenue
Female	\$75,191
Male	\$157,890

2. High-Spending Discount Users

839 customers used a discount code and still spent above the dataset's average purchase amount (\$59.76) — showing discounts aren't only attracting price-sensitive, low-spend shoppers.

3. Top 5 Products by Average Rating

Item	Average Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

4. Shipping Type Comparison

Shipping Type	Avg. Purchase Amount
Standard	\$58.46
Express	\$60.48

5. Subscribers vs. Non-Subscribers

Subscribed	Customers	Avg. Spend	Total Revenue
Yes	1,053	\$59.49	\$62,645
No	2,847	\$59.87	\$170,436

Non-subscribers generate the majority of revenue simply due to volume, but subscribers spend at a nearly identical average — meaning subscription status isn't currently correlated with higher-value baskets. That's an opportunity, not a weakness: converting even a fraction of non-subscribers wouldn't cannibalize spend per order.

6. Discount-Dependent Products

Item	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

7. Customer Segmentation

Segment	Customers
Loyal	3,116
Returning	701
New	83

The customer base skews heavily loyal — over 79% fall into the "Loyal" segment based on purchase history, suggesting the challenge isn't retention but acquisition of genuinely new customers.

8. Top 3 Products per Category

Category	Rank	Item	Total Orders
Accessories	1	Jewelry	171
Accessories	2	Sunglasses	161
Accessories	3	Belt	161
Clothing	1	Blouse	171
Clothing	2	Pants	171
Clothing	3	Shirt	169
Footwear	1	Sandals	160
Footwear	2	Shoes	150
Footwear	3	Sneakers	145
Outerwear	1	Jacket	163
Outerwear	2	Coat	161

9. Repeat Buyers & Subscriptions

Subscribed	Repeat Buyers (>5 purchases)
No	2,518

Repeat buying behavior doesn't require a subscription — most repeat buyers aren't subscribed at all, reinforcing that subscription and loyalty are currently two separate dynamics in this customer base.

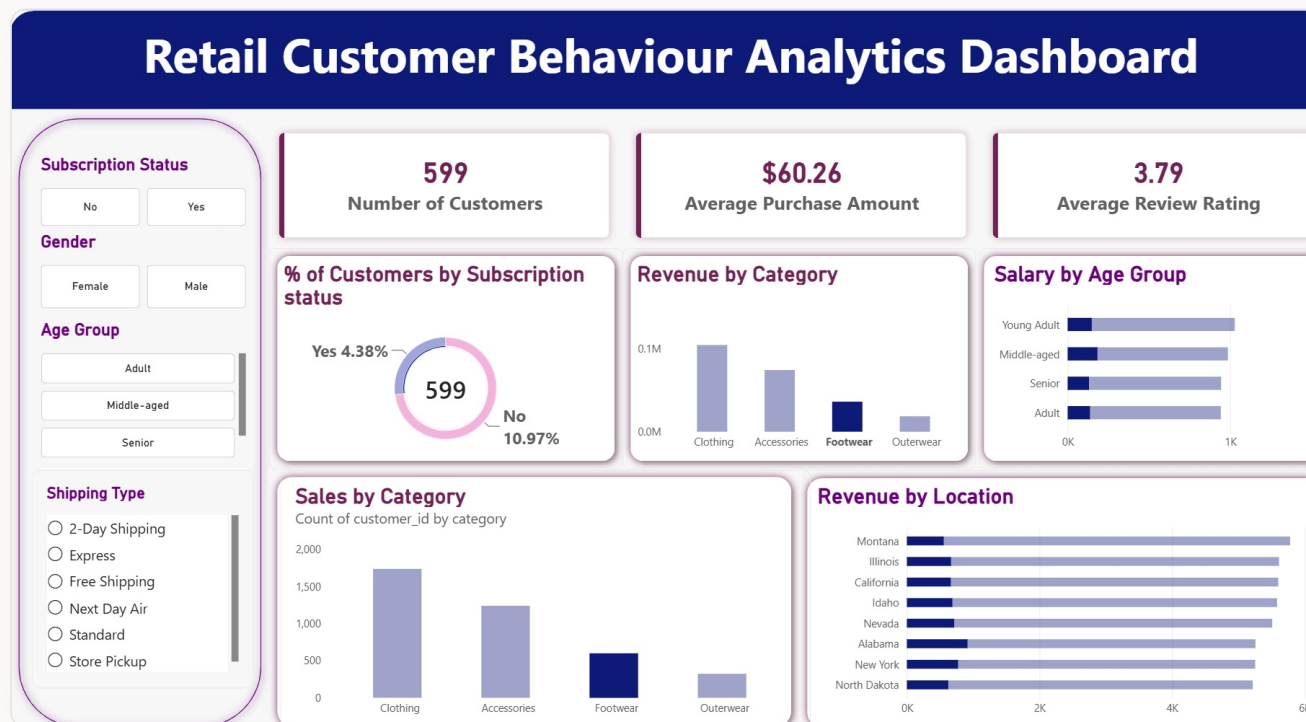
10. Revenue by Age Group

Age Group	Total Revenue
Young Adult	\$62,143
Middle-aged	\$59,197
Adult	\$55,978
Senior	\$55,763

Revenue is fairly evenly spread across age groups, with Young Adults contributing marginally the most — there's no single age group carrying the business disproportionately.

5. Dashboard in Power BI

The cleaned data was connected to Power BI (Import mode via PostgreSQL) to build an interactive dashboard, with filters for subscription status, gender, age group, and shipping type.



6. Business Recommendations

- **Boost subscriptions** — since subscribers and non-subscribers spend at similar averages,

promote exclusive subscriber benefits to convert high-frequency non-subscribers rather than relying on discounting alone.

- **Invest in customer loyalty programs** — reward repeat buyers to move more of the "Returning" segment into "Loyal," and study what differentiates the small "New" segment to improve acquisition.
- **Review discount policy** — products like Hats and Sneakers rely on discounts for roughly half of their sales; assess whether margin is being eroded unnecessarily on items that would sell at full price.
- **Lead with top-rated products in campaigns** — Gloves, Sandals, and Boots have the highest average ratings and are natural candidates for featured placement or upsell bundles.
- **Target high-revenue segments** — Young Adults and Express-shipping users show slightly higher spend; tailor campaigns and shipping incentives toward these groups first.